

Refrigeration:

It is defined as the process of reducing the temperature of a substance below that of the surrounding atmosphere and maintaining this lower temperature within the boundary of a given space.

The machine employed to produce the refrigeration effect is called refrigerator.

Application of refrigeration

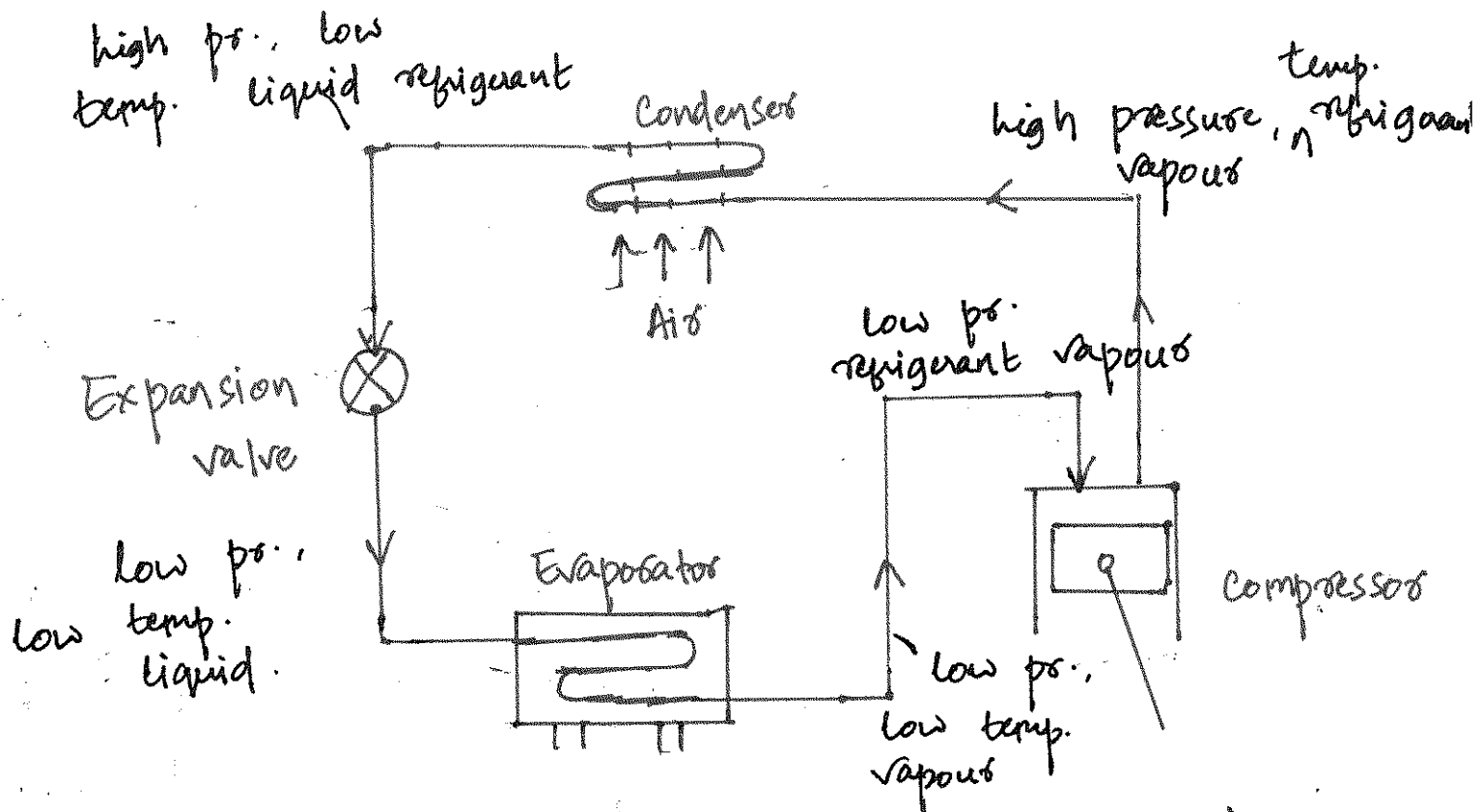
- Manufacture of ice
- Processing of food products & beverages.
- Preservation of perishables like food, drugs etc.
- Industrial air conditioning
- Comfort air conditioning of hospitals, hotels, theatres, residence etc.
- Chemical and related industries
- Liquefaction of gases such as nitrogen, oxygen, natural gas etc.
- Manufacturing & treatment of metals

TYPES OF REFRIGERATION

- vapour compression refrigeration
- vapour absorption refrigeration

VAPOUR COMPRESSION SYSTEM

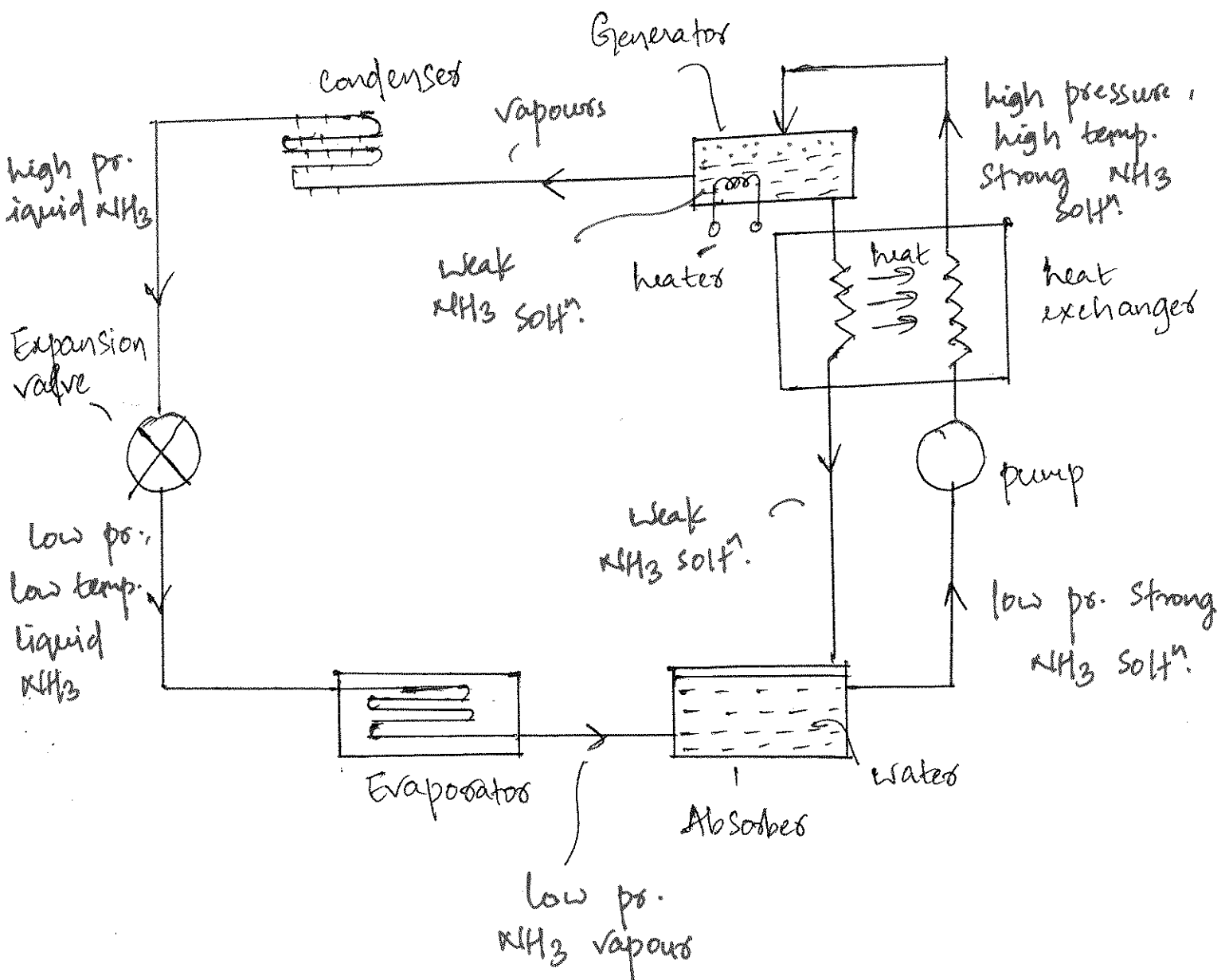
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In this method, the vapour alternately undergoes a change of phase from vapour to liquid and liquid to vapour during the completion of a cycle. The latent heat of vapourisation is used for absorbing the heat from the refrigerated space and the same is rejected during condensation. Four fundamental processes are completed in one cycle. They are compression, condensation, expansion and evaporation. The refrigerant in the form of vapour at low pressure and temperature enters the compressor. In the compressor, it is compressed isentropically and subsequently its pressure and temp. increases.

this compressed vapour then enters the condenser. In the condenser, heat is transferred from the hot refrigerant vapour to air. liquid refrigerant passes to expansion valve and its pressure reduces. It finally passes on to the evaporator where it absorbs heat from the cabinet / space to be cooled. It acquires low pressure, low temp. vapour state. the cycle is repeated.

Vapour absorption refrigeration system



Vapour absorption refrigerator is a heat operated system. Here the compressor is replaced by an absorber, generator and pump. Refrigerant is highly soluble in solution known as absorbent. The liquid refrigerant (NH_3) in the evaporator, absorbs the heat from the medium to be cooled and it becomes vapour. The low pressure vapour then passes to absorber. Here the ammonia vapour is dissolved in weak NH_3 solution or water, producing strong NH_3 solution. The strong solution is then pumped to a generator through the heat exchanger at high pressure. While passing through the heat exchanger, the strong ammonia solution is warmed up by the hot weak NH_3 solution flowing from the generator to the absorber. The weak strong NH_3 solution is heated by an external source in the generator. Due to heating, the vapour gets separated from the solution. The vapour which is at high pressure and high temp. is condensed to low temp. in condenser by cold water circulation. The high pressure liquid ammonia then passes through the expansion valve where it expands to low pressure & temp. The low pressure - low temp. NH_3 liquid again enters the evaporator.

where it absorbs heat from the room and the cycle repeats.

Comparison between vapour absorption & vapour compression system

Vapour absorption

- Works on heat energy
- Energy required to run the system is less
- No moving parts, so operation is quite.
- Due to absence of moving parts, maintenance cost is less.
- Capacity is beyond 1000 tons
- At reduced loads, it is efficient
- Compact in size
- No chance of refrigerant leakage
- Refrigerant used is Ammonia (NH_3)

Vapour compression

- Works on mechanical energy.
- It is more because of compressors.
- Noisy due to presence of compressor.
- Maintenance cost is more.
- Max. Capacity is 1000 tons.
- Performance decreases with increase in load
- Bulky in size
- More chance of leakage of refrigerant due to wear of reciprocating parts of compressor.
- Freon-12 is used as refrigerant.

Refrigerant

It is a medium of heat transfer which absorbs heat by evaporating at a low temp. and gives up heat by condensing at high temp and pressure.

Properties of a refrigerant

1) Thermodynamic property

- i) Low boiling point
- ii) Low freezing point
- iii) high critical temperature
- iv) high latent heat of vapourization

2) Physical properties

- i) Low specific heat
- ii) Low specific volume of vapour
- iii) high thermal conductivity
- iv) Low viscosity
- v) high electrical insulation

3) Safe working temp.

- i) Non toxic
- ii) Non flammable and non explosive
- iii) Non-corrosive

4) Other properties

- i) high coefficient of performance
- ii) Availability & cost
- iii) Ease of locating leakage
- iv) Ease of handling
- v) Low toxicity and flammability.

Types of refrigerants

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- * Ammonia (NH_3) - in ice plants
- * Carbon dioxide (CO_2) - marine refrigerators
- * Methyl chloride (CH_3Cl) - domestic and industrial refrigerators
- * Dichloro difluoro methane) - domestic refrigerators, water coolers.
- * Freon 12 - domestic refrigerators, cold storage,
- * Freon 22 - Air conditioning plants, food freezing & storage.

(Chloro difluoro methane)

AIR - CONDITIONING

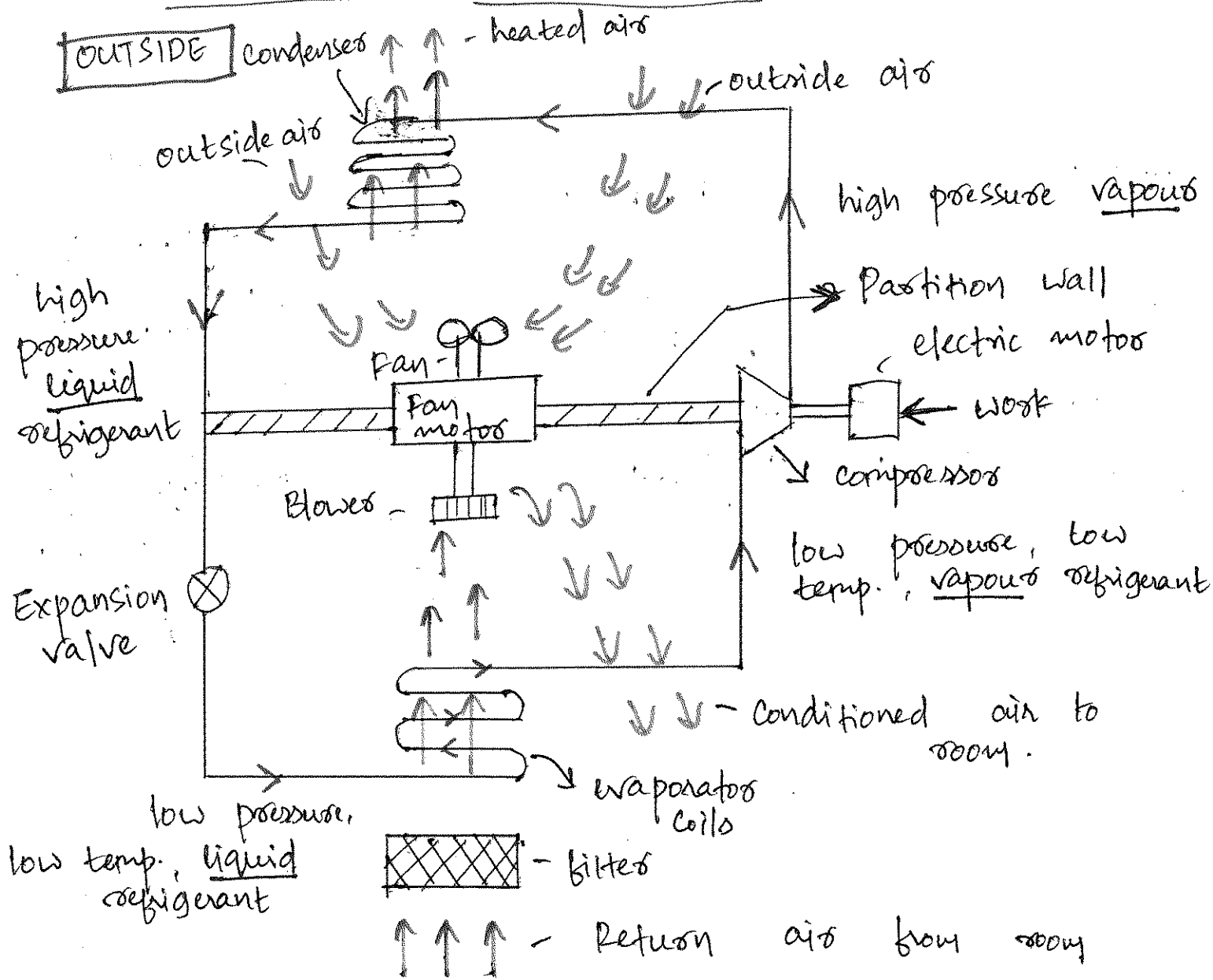
It is process which cools, cleans and circulates air and controls the moisture content. It changes the condition of air in an enclosed area.

Air conditioning is divided into comfort and industrial air-conditioning. The controlled atmosphere which gives maximum comfort to human beings is known as comfort air-conditioning. It is used in houses, offices, shops, restaurants, theaters, hospitals etc.

the conditioned atmosphere which is required in the processing or manufacturing plant is known as industrial air conditioning. It finds application in food processing, photographic, textile, paper and machine tool industries.

WINDOW TYPE ROOM AIR CONDITIONER

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ROOM SIDE

the unit consists of a vapour compression refrigeration system, blower and double shaft motor that drives a fan at one end and blower at other end. the room side and outside side is separated by an insulated partition wall. Blower sucks the warm air from the room through the air filter and the evaporator. the refrigerant inside the evaporator coils gets vaporized by

absorbing the heat from the room.
Motor runs the blower to deliver the cooled air into the room.

the refrigerant vapour from the evaporator is compressed in the compressor to high pressure and temperature.

the high pressure vapour enters the condenser where it is cooled by the outside air which is circulated by fan.

the high pressure refrigerant is reduced to low pressure by passing it through an expansion valve. the low pressure and temperature refrigerant again enters the evaporator where it absorbs the heat from the room and cycle repeats.

Desired temperature in the room is controlled by controlling the operation of compressors with the help of thermostat.

Central air conditioning System

It is suitable for air-conditioning large spaces such as theatres, restaurants, exhibition halls, public buildings etc. It accomplishes the following:-

- Eliminate dust, soot and germs from the air entering the intake ducts.
- Warm the air in winter
- Increase the humidity in dry condition
- Circulate the conditioned air evenly.

Comparison between refrigeration and air conditioning systems

Refrigeration

- * Maintains temp. of space well below surrounding
- * unit is located completely in the room
- * Fan or blower is not required
- * capacity is high
- * Heavy weight
- * used to preserve perishables

Air conditioning

- * Maintains temp., cleanliness, air-flow rate & humidity.
- * located at the window of the room
- * It is required
- * It is low
- * light in weight
- * used to provide congenial atmosphere for human comfort.

~~Advantages of air conditioning~~

- Reduced